Green Building Observership Report

Kanidi Biotech LLP, Greater Noida

27 March 2025

Prepared by AADVWAYA PASSI

Table of Contents

Introduction 2

About the IGBC Rating 2

How the Plant Got the Rating 2

Main Green Features 2

- Energy 2
- Water 2
- Building and Staff Facilities 3

Compulsory Requirements 3

Results 3

What I Learned 4

Conclusion 4

Introduction

I visited Kanidi Biotech LLP in Greater Noida to understand how the plant got its green building certification. The company makes cosmetic products, including organic cosmetics. During my visit, I learned about the different steps taken by the plant to save energy and water and to make the workplace better for the staff.

About the IGBC Rating

The Indian Green Building Council gives ratings to buildings and factories that follow environment-friendly practices. Kanidi Biotech LLP received the IGBC Green Factory Building Platinum rating in 2023. The plant scored 81 points out of 100, which is the highest rating level.

How the Plant Got the Rating

- The project was registered with IGBC.
- A team was made to work on the green building requirements.
- The plant checked all the compulsory requirements before applying.
- Drawings, calculations and other details were submitted to IGBC.
- After the final review and virtual audit, the plant received the Platinum rating.



Solar panels on the rooftop of factory

Main Green Features

Energy

- A 182 kW solar plant has been installed on the rooftop. It gives around 45.5% of the electricity used for non-production work.
- PNG is used for the generator sets and boiler. Biogas is also made from organic waste.
- Separate meters are used to check electricity used for lighting, air-conditioning, pumping and production.

Water

- Rainwater is collected in percolation pits. The pits have a capacity of 180 cubic metres, while the site runoff is 144 cubic metres.
- The plant has sewage and effluent treatment plants. Treated water is used again for flushing, gardening and cooling.
- Self-closing taps, low-flow fixtures and flow meters help reduce water use.
- An oil trap is used for utensil-washing water before it goes into the drain.

Building and Staff Facilities

- Light-coloured roof tiles help keep the building cooler.
- Outdoor lights point downwards so that light does not spread outside the plant.
- Ramps and accessible washrooms are available for people with disabilities.
- There are break areas, a gym, a no-smoking policy and EV charging points for staff.



Compulsory Requirements

Before getting points, the plant had to complete the basic IGBC requirements. Some of these are shown below.

Prerequisite	How the plant satisfied it
Compliance with local regulations	Fire NOC, CPCB approvals and a project-head declaration submitted.
Soil erosion & sedimentation control	Sedimentation system was planned and no soil was eroded in the drains. Topsoil and vegetation was preserved during construction.
Rainwater harvesting (50% run-off)	Percolation pits of 180 cu.m capacity capturing 100% of the 144 cu.m site run-off.
Low-flow water fixtures	Low - flow WCs, urinals, showers and faucets giving about 50% water saving.
CFC-free equipment	VRF air-conditioning using R-410A, free of CFCs.
Minimum energy performance	Verified by whole-building energy simulation against the ASHRAE 90.1 baseline.
Handling of non-process waste	Segregated bins for Different and Multiple waste streams have been placed across the plant and a central waste-segregation facility is in place.
Tobacco-smoke control	Whole campus declared no-smoking, with permanent signage and policy.
Minimum fresh-air ventilation	Ventilation 30% above the IGBC requirement for conditioned and ventilated spaces.
Asbestos-free building	Owner declaration confirming no asbestos-based materials were used.

Results

The plant saves energy through solar power and uses less water through low-flow fittings and reuse of treated water. It also gets daylight in many working areas, manages waste separately and provides a healthier workplace for employees.

Area	Outcome
Energy	51% energy-cost savings vs the ASHRAE 90.1-2004 baseline; 182 kW rooftop solar meeting ~45.5% of non-process energy.
Water	~50% lower water use via low-flow and pressmatic fixtures; 100% of run-off harvested; sewage treated and reused on site.
Daylight & air	Daylight in 84.5% of occupied areas; fresh air 30% above requirement.
Materials	96% of construction waste diverted from landfill; 81% of materials sourced within 500 km; 89% rapidly renewable wood.
Emissions	PNG fuel for gensets and boiler, solar generation and biogas reduce fossil-fuel use and carbon.
Welfare	Accessible, daylit, no-smoking, asbestos-free workplace with break-out spaces and a gym.

What I Learned

I learned that a green factory is not only about planting trees or using solar panels. It also includes saving water, managing waste, using good building materials and looking after the people who work there. I also learned that getting a green building rating needs proper planning and proof for every step.

Conclusion

This visit helped me understand how Kanidi Biotech LLP uses practical methods to reduce waste and save natural resources. The Platinum rating shows the effort made by the plant in energy saving, water reuse and staff welfare.